Week 3: Literature Search & Database Thinking

Reading Research Intros, Boolean Logic, and Testing Ideas

Instructor / Mentor

Kiki Research Training Program

Week 3

Contents

Guided Reading Workshop

Today's Class = A Live Workshop

What we do together in class

1. Read the introduction of the assigned MIT master's thesis together.
2. Learn how to extract the **problem**, **gap**, **keywords**, and **claims**.
3. Use that introduction as a launch point for **Google Scholar** searches.
4. Upgrade a naive search into a **Boolean search strategy**.
5. Connect literature review to **testable research design** and **database thinking**.

Core mindset

A strong literature review is not passive reading. It is an active process of **mapping a field**, **finding gaps**, and **preparing the next test**.

How Should We Read an Introduction?

- **Problem:** What real-world or scientific issue is the paper trying to solve?
- **Motivation:** Why does this matter now?
- **Gap:** What is still missing in the existing literature?
- **Claim:** What does the author say they contribute?
- **Method hint:** What type of evidence or experiment will probably come later?

Never read like a highlighter machine

Do not underline everything.

- Circle the **main question**
- Box the **keywords**
- Mark the **gap sentence**
- Predict the **test or evaluation**

Live Reading Template: The MIT Thesis Introduction

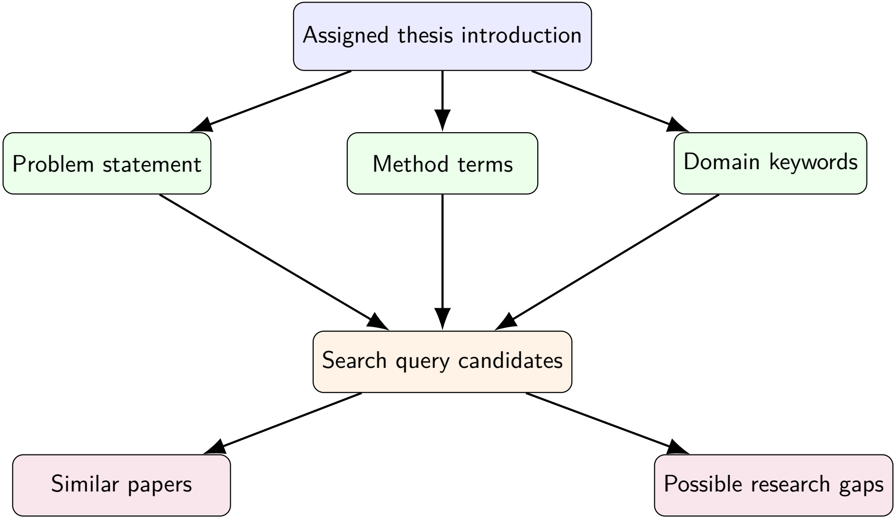
While reading together, ask these 4 questions

1. **What system are we studying?** Example: power grid, energy market, smart meter, EV charging
2. **What exact pain point appears?** Example: instability, forecasting error, coordination cost
3. **Which nouns repeat?** These repeated nouns become our search keywords.
4. **How will the author prove the idea?** Simulation, experiment, case study, comparison, benchmark?

Instructor move during class

Read paragraph by paragraph. After each paragraph, stop and force the class to summarize it in **one sentence only**. If they cannot do that, they did not really digest it.

From One Introduction to a Search Map



Search Strategy & Boolean Logic

Naive Search vs. Research Search

Naive search

```
smart grid AI
```

- Too broad
- Mixed quality results
- Hard to reproduce
- Weak connection to a precise question

Research search

```
("smart grid" OR "power system")
AND
("load forecasting" OR demand
response) AND
(residential OR household)
```

- Narrower and intentional
- Easier to justify
- Easier to refine and document
- Closer to the literature gap

A Practical Google Scholar Workflow

1. Start with **one seed paper** from the assigned reading.
2. Extract 5–8 strong keywords from the introduction.
3. Run one broad search and inspect titles for terminology.
4. Use quotation marks for exact phrases: `"demand response"`.
5. Add **AND / OR / NOT** to control the result set.
6. Filter by year when the field moves fast.
7. Open the most relevant papers, then use **cited by** and **related articles**.
8. Save a mini trail: query used, why opened, why kept or rejected.

What students should learn

Search is not magic. It is an **iterative experiment**.

NSF vs. Google Scholar: Two Different Search Jobs

NSF search = a sharper frontier lens

- Shows **funded projects** and active directions.
- Good for seeing what topics agencies think are promising.
- Best for asking: **where is research money moving now?**

Google Scholar = a broader literature map

- Shows **papers, citations, and related articles**.
- Better for building a comprehensive reading list.
- Best for asking: **what has the literature already said?**

Practical strategy

Start with **NSF** for frontier signals, then use **Google Scholar** to expand and verify the literature map.

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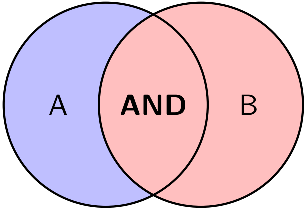
Boolean Logic = A Filter on Sets

- **AND**: keep only items in *both* sets.
- **OR**: keep items in *either* set.
- **NOT**: remove items from a set.



Search meaning

`solar AND battery`
results must discuss both topics.
`microgrid OR "distributed energy resources"`
results may use either vocabulary.
`forecasting NOT traffic`
remove an unwanted topic.

A	B	$A \wedge B$
0	0	0
0	1	0
1	0	0
1	1	1

How a Query Becomes Better, Step by Step

Step 1: Too vague

```
smart grid
```

Step 2: Add the task

```
"smart grid" "load forecasting"
```

Step 3: Add synonyms with OR

```
("smart grid" OR "power system") AND ("load forecasting" OR prediction)
```

Step 4: Add scope and remove noise

```
("smart grid" OR "power system") AND ("load forecasting" OR prediction)
AND (residential OR household) NOT traffic
```

Boolean Logic Also Appears in SQL

Search query logic:

`("smart meter" OR AMI) AND outage NOT review`

Database filter logic:

```
WHERE (keyword = 'smart meter'
       OR keyword = 'AMI')
  AND topic = 'outage'
  AND doc_type <> 'review'
```

Big idea

Boolean logic is not just a math topic. It is the language of:

- search engines,
- databases,
- programming conditions,
- and even research screening rules.

From Literature to Research Tests

A Paper Is Not Just to Be Read — It Must Be Tested

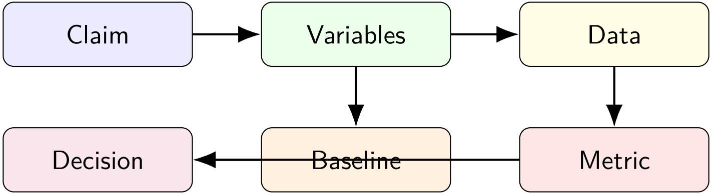
After reading the introduction, ask: what needs to be verified?

1. What is the main claim?
2. Which variables matter?
3. What data would we need?
4. What baseline or comparison is fair?
5. What metric tells us whether the idea is actually better?

Example pattern

Claim: "Method A improves household load forecasting."
Test: Compare Method A against a baseline on the same dataset using MAE / RMSE.

Turning an Introduction into a Testable Design

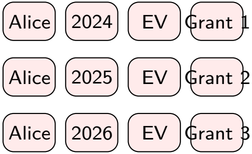


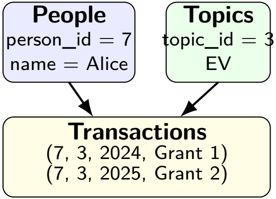
Research habit

A good literature review should make you ask: **What would count as evidence?**

Why Databases Matter

The Box Game: Why a Database Beats a Naive Table





Naive table / spreadsheet

Many little boxes, but the same word must be written again and again.

Database design

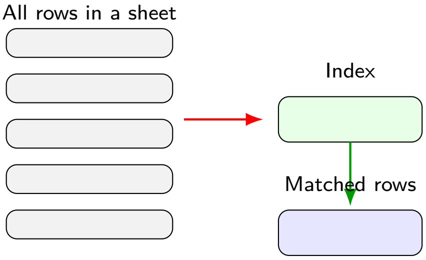
Store repeated facts once, then connect them with keys.

Essential reason

Databases separate **entities** from **transactions**: repeated facts are stored once, updates are safer, and search works on organized structure instead of messy repetition.

Why Database Search Is Faster and More Reliable

1. **Less repetition:** one fact is stored once instead of copied across many cells.
2. **Keys and relations:** records are connected by IDs, so joins are systematic.
3. **Indexes:** the system builds lookup paths, like a catalog, so it does not read every box.
4. **Query engine:** the database chooses an efficient plan for filtering, sorting, and combining tables.
5. **Consistency rules:** constraints reduce contradictions and broken references.



Box-game image

Spreadsheet search often means checking box after box. Database search can jump through an **index** to the right boxes first.

Big-Data Intuition: Why Databases Become Necessary

- **Example 1: Smart meter data**
  1,000,000 meters $\times$ 96 readings per day $=$ 96,000,000 rows per day.
- **Example 2: Literature tracking**
  Hundreds of papers quickly become thousands of rows once we store title, author, year, keywords, method, dataset, and notes.
- **Example 3: Experiment logs**
  Each run may have model version, hyperparameters, timestamp, dataset split, and metrics.

The point

A database is not "just another table." It is a system designed to **store, index, retrieve, and combine** massive structured data efficiently.

Database Thinking: Tables, Keys, and Queries

- A **table** stores one type of entity.
- A **row** is one record.
- A **column** is one attribute.
- A **key** helps us identify or connect records without rewriting the same fact everywhere.
- A **query** asks the database to return only what we need, using structure built underneath.

Library analogy

A database is like a giant library with:

- shelves = tables
- books = rows
- labels = columns
- catalog rules = indexes / keys

Bottom-level idea

The power is not only in storing data. The real advantage comes from **data organization**: normalized structure, indexes, and query planning make retrieval scalable.

SQL Teaser: The Basic Shape of a Query

```sql
SELECT title, year, method
FROM papers
WHERE topic = 'load forecasting'
  AND year >= 2022
  AND dataset = 'residential'
ORDER BY year DESC;
```

- SELECT: which columns do we want?
- FROM: which table do we read?
- WHERE: which rows satisfy our Boolean conditions?
- ORDER BY: how should results be sorted?

Workshop Tasks & Homework

In-Class Workshop Tasks

Task 1: Guided reading

Annotate the MIT thesis introduction with **problem / gap / keywords / expected test**.

Task 2: Search refinement

Start from one naive query, then improve it three times using quotation marks and Boolean logic.

Task 3: Testing mindset

For one claim from the reading, propose: dataset, baseline, metric, and what result would count as evidence.

Homework for This Week

1. Do your own literature search on the topic **smart grid**.
2. Keep a short search trail: at least **three queries** you tried, and how each revision improved the results.
3. Select **three papers** that you found interesting.
4. Prepare a short presentation for next class: **no more than 7 slides in total**.
5. In the presentation, introduce the **three papers** and explain **why each one is interesting** to you. (In other words, you should be able to explain why others are not so interesting to you. Why do these 3 papers win?)

Thank you!